

FINM 25000 Homework 2: Technical Indicators & Strategy Backtesting

Ticker: AAPL

Period: 5 years of daily OHLCV bars

Data source: Synthetic mock data

Initial capital: \$100,000

Constraints: long-only, no leverage, no short selling

Artifact note:

These artifacts were generated in mock mode because Alpaca credentials were not available in this local repo at generation time.

Backend detail: Missing Alpaca credentials. Copy .env.example to .env and set ALPACA_API_KEY and ALPACA_SECRET_KEY.

Strategy Descriptions

Buy & Hold: fully invested benchmark used as the baseline comparison.

Trend Following: buys when MACD is above its signal line, ADX is above 25, and price is above the 50-day SMA. It exits when MACD crosses below its signal or price loses the 50-day SMA.

Mean Reversion: buys oversold conditions where RSI is below 30 and price is below the lower Bollinger Band. It exits when RSI is above 70 or price moves above the upper Bollinger Band.

Custom Strategy: combines trend, momentum, volatility, and volume. It requires $EMA_{20} > SMA_{50}$, $MACD > signal$, RSI between 45 and 70, price above the Bollinger midline, positive CMF, and a normal ATR regime. It exits on trend, momentum, or volume deterioration.

Performance Comparison

Strategy	Total Return	CAGR	Volatility	Sharpe	Sortino	Max DD
Win Rate						
Buy & Hold	26.0%	4.7%	21.3%	0.32	0.55	-38.3%
100.0%						
Trend Following	13.2%	2.5%	12.3%	0.26	0.26	-16.6%
34.9%						
Mean Reversion	73.0%	11.6%	14.7%	0.82	0.97	-21.8%
70.0%						
Custom	6.6%	1.3%	13.2%	0.16	0.17	-19.7%
31.6%						

Discussion

The highest Sharpe ratio in this run belongs to Mean Reversion. The key comparison is risk-adjusted performance, not just raw return, so the

equity curve and drawdown chart should be reviewed together with the Sharpe, Sortino, and maximum drawdown metrics.

Charts

The repository includes HTML chart exports in the charts/ folder:

- price chart with indicators and buy/sell signals
- equity curve comparison
- drawdown comparison